



Integrating degrowth and efficiency perspectives enables an emission-neutral food system by 2100

Benjamin Leon Bodirsky^{1,2,5}, David Meng-Chuen Chen^{1,3,5}✉, Isabelle Weindl¹, Bjoern Soergel¹, Felicitas Beier^{1,3}, Edna J. Molina Bacca^{1,3}, Franziska Gaupp^{1,4}, Alexander Popp¹ and Hermann Lotze-Campen¹

Degrowth proponents advocate reducing ecologically destructive forms of production and resource throughput in wealthy economies to achieve environmental goals, while transforming production to focus on human well-being. Here we present a quantitative model to test degrowth principles in the food and land system. Our results confirm that reducing and redistributing income alone, within current development paradigms, leads to limited greenhouse gas (GHG) emission mitigation from agriculture and land-use change, as the nutrition transition towards unsustainable diets already occurs at relatively low income levels. Instead, we show that a structural, qualitative food system transformation can achieve a steady-state food system economy that is net GHG-neutral by 2100 while improving nutritional outcomes. This sustainable transformation reduces material throughput via a convergence towards a needs-based food system, is enabled by a more equitable income distribution and includes efficient resource allocation through the pricing of GHG emissions as a complementary strategy. It thereby integrates degrowth and efficiency perspectives.

Proponents of degrowth propose an ‘equitable downscaling of production and consumption that increases human well-being and enhances ecological conditions at the local and global level’, to be undertaken voluntarily and democratically¹. Degrowth proponents see this transformation as necessary to meet environmental goals, such as those laid out in the Paris Accord or the Planetary Boundaries², as degrowth would reduce demand-driven pressures on the environment³. Degrowth arguments also include a strong environmental and economic justice component: downscaling and redistribution of the high-income economies of the Global North would allow for ‘more space’ for countries of the Global South to develop and still grow within environmental limits^{4,5}. A reduction of gross domestic product (GDP) is generally not seen as an objective per se, but rather as the expected consequence of the downscaling of material and energy throughput⁴. Some practitioners may also target a reduced GDP as a means, given the strong association between GDP and certain ecological impacts⁶.

While the theoretical debate around degrowth is emergent and lively^{4,7,8}, few studies have attempted to quantify degrowth proposals in modelling work to assess how effectively such proposals meet stated goals. Integrated assessment modelling of future environmental scenarios without the precondition of economic growth, especially in the Global North, are needed^{3,9}.

Degrowth scenarios have been analysed recently on a macroeconomic scale and for the energy system^{10,11}, but not for the land and food system. A focus on the implications for the food and land system is long overdue because this is today the largest cause of biodiversity and ecosystem destruction^{12,13} and because it emits one-third of global greenhouse gas (GHG) emissions along the entire supply chain¹⁴. Furthermore, land-use and land-use change (LULUC) emissions may alone preclude achieving the 1.5 and 2 °C climate targets as decided upon in the Paris Accord¹⁵.

An assessment of degrowth scenarios for the land and food system should aim to cover the breadth of degrowth proposals and decompose the main dynamics that lead to GHG reduction and reduced economic activity. A review¹⁶ structures the degrowth literature according to three economic policy objectives: (1) a sustainable scale of the economy, keeping the economy within the planetary boundaries of a ‘safe operating safe’ with pollution remaining within environmental absorption capacity²; (2) ‘fair redistribution’ between rich and poor, on both national and international scales; and (3) ‘efficient allocation’ of resources to maximize well-being¹⁷.

In this article we use a quantitative land system model¹⁸ to decompose different dynamics behind a degrowth pathway and scrutinize the consequences of different viewpoints in the debate by constructing a range of scenarios, aligning with the policy objectives identified above.

Results

Scenario set. Our set of exploratory scenarios of stylized socioeconomic transformations covers different constituents of both degrowth proposals and efficiency-based approaches already theorized and modelled (Table 1). Two scenarios of ex ante GDP reduction (GDP-CAP) or redistribution (GDP-FAIR) aim at discussing the potential and limitations of purposefully reducing or redistributing GDP, as opposed to the following scenarios, which address qualitative aspects of the food system with no prior assumptions of GDP—in line with current degrowth scholarship⁴. The food-preference-change scenario (DIET) is oriented along a needs-based provision of food; the efficient-allocation scenario (EFF) assumes a global GHG tax internalizing the costs of climate change stemming from food supply chains. Finally, the sustainable scenario combines GDP-FAIR, DIET and EFF.

¹Potsdam Institute for Climate Impact Research (PIK), Member of the Leibniz Association, Potsdam, Germany. ²World Vegetable Center, Tainan, Taiwan.

³Humboldt-Universität zu Berlin, Berlin, Germany. ⁴EAT, Oslo, Norway. ⁵These authors contributed equally: Benjamin Leon Bodirsky, David Meng-Chuen Chen.

✉e-mail: david.chen@pik-potsdam.de

Table 1 | Scenario descriptions, decomposing proposals in the degrowth literature

Scenario	Scenario settings	Literature analogue
Baseline (BAU)	Baseline scenario along middle-of-the road development (SSP2 storyline), including moderate economic and population growth ²⁹	Default baseline scenario of many integrated assessment analyses of the land and food system ^{21,70}
Capped income (GDP-CAP)	Total per-capita income decreases by 2030 to a limit of US\$12,746 (05PPP), which is the World Bank threshold between middle- and high-income countries ⁷¹ . Countries below the threshold follow their normal growth path. See Supplementary Fig. 1 for income trajectories of scenarios	The selected threshold corresponds roughly to the income level where mean life satisfaction, life expectancy, infant mortality and participation in education begin to saturate ^{35,72,73,76} . This scenario does not include a qualitative transformation but rather simulates income reduction as measured by GDP, within the current economic paradigm
Fair redistribution (GDP-FAIR)	Like GDP-CAP, but increasing per-capita income to US\$12,746 (05PPP) for countries below this limit by 2030. In both FAIR and GDP-CAP, the historically observed relation between dietary patterns and income is maintained	A fair and equitable economy in terms of income, where countries of the Global South have increased GDP per capita, given redistribution from higher-income countries ^{4,77} . This scenario does not include a qualitative transformation but rather simulates income redistribution within the current economic paradigm
Preference change (DIET)	A dietary shift towards the planetary health diet ³⁶ by 2030, which sees much lower animal-source food consumption, combined with a reduction of food waste ⁵⁶	A shift in diet has been proposed as part of any degrowth process in the agrifood sector, especially in a way that meets human needs for a good life, while remaining within planetary boundaries ⁷⁸ . Such a shift should result in reduced consumption of emissions-intensive products ⁴
Efficient allocation (EFF)	GHG tax in line with the Paris Accord goal of 1.5 °C warming. Non-CO ₂ GHGs are priced according to their CO ₂ -equivalent warming potential	An efficiency-based mitigation approach, reducing emission sources at the same marginal mitigation costs across emission sources by technical mitigation measures or relocation of resources and production factors. Achieved via a GHG price that internalizes externality costs of pollution into production costs ³¹ . Emission pricing is the key instrument in most integrated assessment analyses of the land and food system ²⁴
Sustainable transformation (TRANS)	Combines GDP-FAIR, EFF and DIET	A sustainability transformation combining societal change and efficient taxing, as a combination of degrowth and efficiency perspectives ³⁰

05PPP, purchasing power parity equivalents in constant 2005 US\$.

The scenario assumptions serve as input to MAGPIE4, a modular open-source framework for modelling global land systems and their environmental impacts¹⁸ (Methods). Our scenarios focus exemplarily on GHG emissions as a central indicator for environmental sustainability. Further environmental indicators such as biodiversity intactness¹⁹, nitrogen pollution²⁰ and water withdrawals have been shown to respond similarly^{21,22}.

There is extensive literature on the process of reorganizing the economy towards a degrowth-oriented system through democratic and participatory institutions^{7,16}. We assume this a priori in this article and instead focus on the implications of such a transformation for the food and land system.

Model outputs. Average yearly GHG emissions from LULUC from 2020 to 2100 are highest in the SSP2 BAU baseline, at an average of 14.4 GtCO₂e yr⁻¹ in 100 yr global warming potential (CO₂e GWP100) emissions (Fig. 1a), and 16.1 GtCO₂e in the year 2050. While the precise warming effect cannot be derived from the GWP100 metric, a comparable scenario¹⁵ shows that business-as-usual emissions from the LULUC sectors alone would lead to global warming between 1.5 and 2 °C. In the ‘capped income’ (GDP-CAP) and ‘fair redistribution’ (GDP-FAIR) scenarios, CO₂e emissions are reduced only modestly (14.4 GtCO₂e for GDP-CAP and 13.9 GtCO₂e for GDP-FAIR in 2050), as the reduced consumption in high-income countries is countered by a continued (GDP-CAP) or rapid (GDP-FAIR) increase in consumption in low-income countries. In fact, the redistribution in the GDP-FAIR scenario even results in higher food demand and increased emissions relative to the SSP2 baseline during the years when redistribution is implemented. Despite the substantial reduction in overall economic activity (global GDP in GDP-CAP is 44% and in GDP-FAIR is 51% that of BAU by 2050; Supplementary Fig. 1), cumulative emissions from 2020 to 2100 are still in excess of the remaining emission budget for

achieving a 1.5 °C climate target with a probability of 66% (ref. ¹⁵), with 1,024 GtCO₂e (GDP-CAP) and 1,077 GtCO₂e (GDP-FAIR). The limited effect (in terms of a reduction of LULUC GHG emissions) for the GDP-CAP and GDP-FAIR scenarios compared with BAU stems from the fact that the main dynamic of the ‘nutrition transition’—increasing demand for land- and emissions-intensive animal-source products as well as increased food waste—unfolds already at relatively low per-capita income levels²³.

The ‘efficient allocation’ scenario (EFF) results in almost a halving of LULUC-related GHG emissions by 2100, with 9.05 GtCO₂e in 2050. This price signal reduces emissions by incentivizing afforestation and less-polluting management practices, and incentivizing land-sparing and investments in yield-improving technological change²⁴ (Methods). The consumer ‘preference change’ (DIET) scenario as part of a degrowth pathway also leads to substantial emissions mitigation (6.93 GtCO₂e in 2050), due to reduced demand-side pressures on the land system (that is, a shift away from instead of towards animal-source products). Finally, a sustainability transformation scenario (TRANS) where fair redistribution is combined with efficient allocation and preference changes reduces cumulative emissions to 1.18 GtCO₂e in 2050. In this scenario, land becomes available for increased afforestation and regrowth of natural vegetation, resulting in negative CO₂ emissions that compensate for some of the residual N₂O and CH₄ emissions. By 2100, this scenario shows net-zero emissions for the food system.

The scenarios also show very different trends in their agricultural material throughput, defined here as the total demand for crops for food, feed and other purposes (Fig. 1c). Both BAU and EFF show a continued increase, nearly doubling the per-capita footprint by the end of the century. The GDP-FAIR redistribution scenario leads to a slightly faster initial increase due to the acceleration of income-driven dietary changes in low-income countries but converges to the slower-growing value of the GDP-CAP scenario

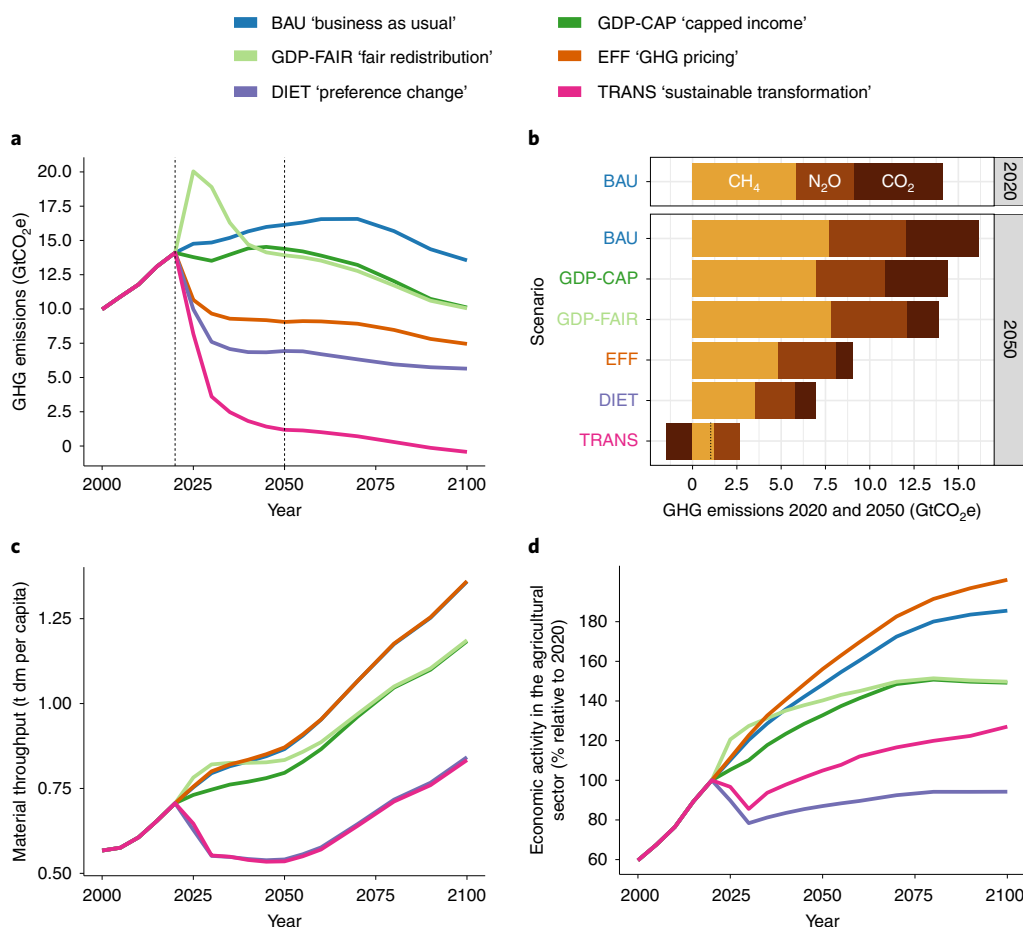


Fig. 1 | Development of emissions, material throughput and economic activity for six scenarios. a, GHG emissions for 2020–2100 in CO₂e GWP100. Dotted lines indicate years of comparison in **b**. **b**, The composition of GHG emissions in 2020 and 2050 as a stacked bar plot for CH₄, N₂O and CO₂ in GtCO₂e GWP100. Dotted line in TRANS shows net total. **c**, Cumulative material throughput of food consumption, calculated as the demand for crop products in tonnes of dry matter (t dm) per capita. **d**, Percentage change of the economic activity in the agricultural sector (expressed as the factor costs of production at constant factor prices excluding land rents, water rents and emission taxes) as a line chart taking the 2020 value as 100%.

in the longer term. Finally, the DIET and TRANS scenarios project a strong initial reduction of the agricultural material throughput, and would only return to a per-capita footprint comparable to today towards the end of the century. Comparing material throughput with GHG emissions, there is no direct coupling between the two quantities in our scenarios. In the EFF and TRANS scenarios, we observe a partial but not complete decoupling of emissions from the material throughput; residual non-CO₂ GHG emissions from agricultural production can be reduced but not completely avoided.

Changes in the material scale and efficiency of the agrifood system also impact economic activity, which we here define as the sum of global agricultural production costs, investments into agricultural R&D, and irrigation and land expansion costs. These costs correspond to production factors such as labour, capital and input commodities that are required by the food system from the wider economy. In BAU, agricultural economic activity grows by 48% from 2020 to 2050 (Fig. 1d). When GDP is reduced in GDP-CAP and GDP-FAIR, agricultural activity is still 89% that of BAU in 2050, although GDP in GDP-CAP is 21% that of BAU by 2050 (Supplementary Fig. 1). This can be traced back to food being a necessity good with an income elasticity smaller than unity. The efficient allocation scenario (EFF) with a GHG tax actually increases agricultural economic activity relative to BAU, as the implementation of mitigation measures increases costs, and as GHG-intensive land expansion has to be replaced by the intensification of existing

croplands. In contrast, the structural change of the DIET scenario, in particular the reduction of animal-source foods, leads to a strong initial reduction and subsequent stabilization of economic activity to slightly below current levels. The TRANS scenario, combining GDP-FAIR, EFF and DIET, shows a short-term decline of economic activity, and then a small but steady increase to a level 20% above current levels, also due to continuous population growth.

The dynamics driving the different scenario outcomes are further described in Supplementary Text 1 and Supplementary Data, section 2.

Discussion

Our study applies a set of quantitative scenario simulations to decompose the different constituents of both degrowth- and efficiency-based visions for the agriculture and food system. We find that to reduce global LULUC GHG emissions, both a change of dietary preferences and a change of economic incentives by emission pricing are crucial. A more equitable income distribution between countries only leads to very limited reductions in LULUC emissions if it remains within the current ‘nutrition transition’ paradigm, that is, the historically observed relations between income and dietary patterns^{23,25}.

Our results are in line with the arguments of degrowth proponents, who differentiate sustainable degrowth from economic recession in that degrowth proposals also transform the economy in a

qualitative manner¹. We show that a simple reduction of income in high-income countries, while maintaining the existing food system, is not very effective in mitigating GHGs; most importantly because the food system—in contrast to the energy system²⁶—already produces relatively high emissions at low per-capita incomes¹⁴. The results from the GDP-FAIR scenario even indicate that the marginal pollution per unit of income is higher at lower incomes. Despite a lower global average GDP in the GDP-FAIR scenario (Supplementary Fig. 1), GHG emissions increase relative to the BAU scenario for the period of redistribution. Also, from the public health perspective, there is no sweet spot of economic development within the classical nutrition transition; the decline of undernutrition is paralleled by a simultaneous rise of malnutrition, the so-called ‘double burden of malnutrition’^{23,27}.

The food system thus needs to be transformed both in terms of its material scale and its qualitative structure: resource-intensive and highly polluting industries, such as the livestock industry, would need to be downscaled along with a reduced consumption of animal protein. Other subsectors such as horticulture should in contrast even be expanded due to their role in preventing malnutrition and chronic diseases^{23,28}. Furthermore, farmers need to adopt low-polluting management practices, and supply chains must shift towards low-polluting source materials.

Our study explores two distinct ways to change the qualitative structure of the food system: emission pricing (EFF), which includes the hidden costs of GHG emissions into the decision-making at various stages of the food supply chain; and changing preferences of consumers (DIET), which influences the structure of the economy from the end of the supply chain.

While emission pricing and the resulting efficient allocation of resources has received a lot of attention in the integrated assessment modelling community²⁹, it is less prominent and indeed often criticized in the degrowth literature^{16,30}. We thus consider it separately from the degrowth narrative, although some emission pricing schemes within a degrowth framework have been proposed³¹. Indeed, such schemes warrant further consideration as our model simulations show high mitigation potentials for the scenarios that introduce a price signal for GHGs into the agricultural and land system (EFF and TRANS). A GHG price does not only incentivize efficiency improvements and the implementation of mitigation technologies, as opposed to, for example, subsidies for mitigation measures or more efficient technologies, but a tax or cap-based policy also circumvents rebound effects that can occur from efficiency improvements and passes on the increased costs as mitigation incentive to upstream sectors and consumers. In these upstream sectors, the price signal may lead to further mitigation, which is not, however, included in our assessment. For instance, higher raw material prices may incentivize food loss reduction or the replacement of emission-intensive animal-based ingredients by plant-based alternatives in the food processing sector³². GHG taxation would also translate into higher food prices (although the tax could be redistributed, discussed further on). Under cost-efficient CO₂ prices, the low cost-share of agricultural products in total food prices³³ and observed price elasticities, the resulting food price changes would not, however, have a transformative impact on dietary patterns (Supplementary Text 2). Consumer tax levels to incentivize sustainable diets would thus need to be much higher than the signal that would reach consumers from a uniform emissions tax³⁴, and changing demand via prices would therefore constitute an inefficient alternative to production-side mitigation from an environmental cost–benefit perspective.

However, the potential of demand-side mitigation is substantial, as shown in the DIET scenario, and the pricing of GHG emissions only achieves minor impacts on this front. Transformative dietary change is therefore a key strategy. This approach is in line with the degrowth arguments that consumption should be oriented along

with the satisfaction of basic human needs and provide living standards such that a ‘good life for all’ is possible within environmental limits³⁵. This diet change is not a mere reduction of consumption, but a qualitative change that can be even considered an improvement with respect to the satisfaction of human needs, given the improved nutritional composition of the dietary shift³⁶.

Transformative dietary change, with its large mitigation potential, can also be achieved with different (non-price) mechanisms such as persuasion, access, empowerment or nudging. Research has so far mostly focused on policy interventions to improve dietary health and compiled and evaluated a broad spectrum of interventions that target food preferences and food environments^{37,38}. Mechanisms such as food labelling³⁹, advertisement bans⁴⁰, dietary counselling^{41,42} and public food provision⁴³ have been identified as effective and often cost-efficient alternatives to fiscal instruments. In economic nomenclature, these policies target preferences (represented, for instance, by parameters such as demand elasticities or floor demand shifters) which are usually assumed exogenous and constant. A consumer with the same income and faced with the same market prices will consume different quantities if she lives in a food environment with low marketing for highly processed foods, if the food availability in the canteen is modified, or if her awareness for environmental or health topics is raised. Research on the dynamics and drivers of preference change, on the influence of food environments, and their impacts on elasticities and floor demand shifters⁴⁴ remains underdeveloped but would be the necessary bridge between a more holistic food system approach and standard economic models. The investigation would also be confronted with welfare economics paradoxes with regard to the evaluation of policy interventions, as the same individual may value a bundle of goods differently before and after the intervention⁴⁵. Considering the strongly reduced burden of disease for both poor and rich consumers³⁶, the more modest and diverse consumption patterns of the EAT–Lancet diet could be considered an improvement of aggregate welfare, despite lower willingness to pay. Improved welfare indicators should therefore go beyond the aggregation of the individual willingness to pay, and could, for example, correct for differences in assets between individuals⁴⁶, or include further indicators that are oriented along outcomes of consumption such as health or subjective well-being⁴⁷.

Qualitative transformations of the food system lead to distinct outcomes for material scale, economic activity and income distribution. Our simulations show that the emission-pricing scenario EFF shows a partial decoupling between emissions and material throughput while leaving the material throughput largely unchanged compared to a BAU scenario. In contrast, the DIET scenario reduces environmental pollution along with material throughput. The results also show that reduced or stable economic activity in the agricultural sector is the consequence, and not the cause, of a sustainable food system. Economic contraction in high-income countries as in the GDP-CAP scenario does not lead to a strong reduction in material throughput or emissions in the food system; in contrast, the qualitative transformation of the DIET and TRANS scenarios leads to a strong reduction in economic activity in the agricultural sector during the transformation phase, resulting after 2030 in almost steady-state economic activities comparable to today’s level (Fig. 1c).

While the environmental benefits of TRANS accrue mainly from the EFF and DIET levers, income redistribution from GDP-FAIR is furthermore a necessary but not sufficient condition for a sustainable transformation, as higher incomes will be required in lower-income countries to afford healthy diets, and to pay for the full price of food that includes the environmental externalities. Although the planetary health diet reduces dietary expenses in high-income countries and many middle-income countries, the average costs of US\$2–3 (11PPP) per day would probably exceed available incomes in many

lower-income countries⁴⁸. Higher incomes in lower-income countries would not necessarily require global economic growth but could also be achieved by international redistribution. Furthermore, while the pricing of emissions may have a regressive distributional effect for lower-income populations, this could be reverted effectively via redistributing tax revenue as per-capita dividends and via international transfers⁴⁹. Indeed, in our EFF scenario, the revenues from emissions pricing collected in 2050 amount to US\$3.36 trillion (05PPP) by 2050, and would be available for redistribution or reinvestment in social policies. Emissions taxation with revenue-neutral redistribution has also been implemented in practice, with redistribution improving public support for the tax⁵⁰.

Finally, mitigation activities of course require economic investment, for example, to better monitor and control nitrogen flows on croplands, to invest in better animal waste management facilities or to breed more efficient crops. Our simulation shows, however, that these additional costs are small in comparison to overall production costs, and a degrowth-informed shift from polluting to mitigation activities still results in a quasi-steady-state land economy under a sustainable transformation. In reality, these investment costs may be further offset by the restoration of ecosystem services⁵¹.

We identify four limitations in this study, pointing toward opportunities for further research. First, our model can only alter the income distribution between countries, but not change the income distribution within countries. Missing global data on the intersection between income distribution and diet distribution within countries still impede such an assessment on the global scale. Yet, inequality between countries is still much higher than inequality within countries⁵². Given observed food demand patterns²³, it can be expected that reducing inequality within countries would have qualitatively similar impacts as reducing inequality between countries, reinforcing the dynamics observable in our study. Second, our study is a sectoral study for the food system. Macroeconomic trends such as per-capita income are exogenously prescribed, and the economic activity in the agricultural sector does not further influence the wider macroeconomy. Impacts on employment and wages may be high and require social and structural change policies⁴. Future assessments should investigate in more detail the interactions of the food system with the energy system and the remaining macroeconomy—for example, to derive consistent bioenergy demand from an energy system running similar degrowth or emission pricing trajectories, or to account for agricultural labour costs given degrowth dynamics in the labour market. Moreover, energy emissions account for 21% of today's global food system emissions and stem in similar shares from energy use in agricultural production, food transport, processing, packaging, retail and consumption¹⁴. These emissions are considerably higher in higher-income countries due to longer supply chains and industrialized food processing. Carbon pricing would reduce these emissions via the decarbonization of the energy system⁵³, while diet change would reduce these emissions via a shift away from processed products. Similarly, food waste reduction could drastically lower CH₄ emissions from wastewater and solid waste, which account for a further 9% of food system emissions¹⁴.

Third, our scenarios focus on two central indicators: GHG emissions and economic activity in the agricultural sector. Previous assessments came to similar conclusions also for an extended range of environmental indicators such as nitrogen use, biodiversity loss or freshwater withdrawals. The current global food system could only nourish 3.4 billion people sustainably, but a qualitatively altered food system could nourish up to 10.2 billion people⁵⁴. The favourable impacts of diet change across environmental indicators has also been highlighted^{22,28}. For a comprehensive welfare assessment of different pathways, further societal targets should also be included²².

Finally, while our scenarios intend to aid in scrutinizing the desirability of goals and pathways, this study cannot assess

the psychological, economic, social or political barriers that prevent such pathways. Future research should assess possible policy bundles to achieve such a rapid and radical transformation, and the political economy, political institutions and social movements that could enable such a change⁵⁵.

The food system should be more explicitly considered in the degrowth debate, as the basic need of food has to be fulfilled in any economy, and as the food system is a major source of GHG emissions and other environmental damage. We find that a sustainably transformed food system would align with a substantially reduced agricultural material throughput, thus shrinking the material scale of the global agrifood system. As a consequence, its economic scale may slightly shrink in the near term and would grow more slowly in the longer term. For the food system, our results quantitatively confirm that degrowth has to be qualitatively different from negative economic growth. The reduction of economic activity is not an *a priori* cause of a sustainable food system but the consequence of it; negative growth or redistribution, if conceived of only as reduction and redistribution of GDP within the current economic development paradigm, contributes little to reducing environmental impacts of the food system while it probably reduces the provision of low-polluting goods and services. Indeed, a reduction in macro-level throughput as such does not guarantee a healthy and sustainable food system, for which bottom-up consumption changes are needed. Emission pricing is a complementary strategy that should receive further attention in the degrowth community, as it leads to more efficient resource allocation. The combination of preference change and efficient allocation integrates well into the concept of a sustainability transformation³⁰, and leads to the most stringent reductions of GHG emissions from the food system. International income redistribution, while not substantially reducing emissions by itself, may still be pivotal in its function to enable lower-income people to afford sustainable diets and to compensate for adverse distributional effects of emission pricing.

Methods

Model. We analyse the degrowth scenarios using the MAGPIE4 open-source framework for modelling global land system¹⁸. The model is well documented and available open source (<https://github.com/magpiemodel>). MAGPIE4 combines socioeconomic and biophysical data to simulate spatially explicit land-use scenarios for the 21st century, along with resultant environmental impacts. At its core, the framework is a partial equilibrium economic model, with the objective function of minimizing production and other costs while meeting food, feed and material demand.

The MAGPIE model is driven by a food demand module²³, which estimates final food demand and dietary composition based on population growth, demographic change and per-capita income. It explicitly accounts for changes in caloric requirements by age, sex, physical activity and body height, and estimates body weight, food waste and dietary composition endogenously based on per-capita income. The model has been econometrically parameterized based on past country-level trajectories of body mass distributions and food demand. Food demand is estimated for four product groups: animal-source foods, empty calories (sugar, oil, alcohol), fruits and vegetables, and staples. For the scenarios that assume preference changes, we only use the information on caloric requirements and use exogenous assumptions for dietary composition and food waste⁵⁶, assuming a shift towards the healthy and sustainable planetary health diet as proposed by the EAT-Lancet Commission⁵⁶.

The consumption of animal-based calories requires the cultivation of animal feed crops, estimated based on regional dynamic feed-baskets⁵⁷. Similarly, the processing of empty calories requires primary crop commodities based on conversion efficiencies from FAOSTAT. Per-capita material demand for crops, processed products and livestock is assumed to grow in proportion to food demand. Bioenergy demand is exogenous and the same for all scenarios simulated in this study. The cultivation of crops requires labour and capital inputs, land, nitrogen fertilizer, seed and, in the case of irrigated crops, water. The location of crop and pasture production is spatially explicit on a 0.5° grid clustered to 200 production clusters. Production location is determined based on cost-competitiveness, considering the limited availability of land, irrigation water, organic fertilizers, and the differences in crop yield potentials and transport costs. Yield potential patterns, irrigation water requirements and irrigation water availability are obtained from the dynamic crop and vegetation model LPJmL (ref. ⁵⁴). Moreover, the model can intensify crop production and achieve higher

yields by investments into research and development as well as by scaling up fertilization to higher nutrient demands. Future technological change forecasts in MAGPIE have been estimated and validated based on historic trends and contemporary data^{58,59}. These yield improvements do not take into account any 'system-disrupting' future technologies, such as the production of food without land⁶⁰, and involve correspondingly higher fertilization needs per hectare and corresponding emissions. The costs of additional yield gains were derived on relating past productivity developments to investments in research, technology and infrastructure⁵⁹.

Food can be traded internationally within certain self-sufficiency thresholds⁶¹. To satisfy increases in demand, the model finds an economic equilibrium between expanding croplands and expanding irrigated areas, relocating crop production to higher-yielding areas.

Emissions include CO₂ emissions from land-use change, as well as non-CO₂ emissions from agricultural production. Land-use change emissions are based on spatially explicit cropland and pasture expansion and carbon stocks from the LPJmL model, including vegetation, litter and soil carbon⁶².

When a price on emissions is activated, this incentivizes the preservation of currently existing forests and the planting of new forests⁶³. This afforestation is based on natural vegetation growth curves and carbon densities of regionally native species, as opposed to plantation forest curves⁶³. The maximum carbon density achievable by afforestation is that of natural forests. Moreover, afforestation in our implementation can only take place in grid cells where carbon densities support greater than 20 tC ha⁻¹, a commonly used threshold for forest and non-forest land⁶⁴. Finally, afforestation in the model does not take place in boreal zones because the albedo effect would offset the climate effect of the CO₂ sequestration⁶⁵. This is implemented via an exogenous constraint⁶⁶, but results in similar outcomes as studies that directly include the albedo effect within the optimization⁶⁷.

CH₄ emissions from enteric fermentation are based on feed-baskets⁵⁷ and the Tier II methodology of the IPCC national reporting guidelines⁶⁸. CH₄ emissions of rice are estimated based on the Tier I methodology of the IPCC national reporting guidelines⁶⁸. N₂O emissions are based on a nitrogen-budget model²⁰ that explicitly accounts for manure availability, crop residues, biological fixation and inorganic fertilizer application, and their direct and indirect N₂O emissions based on IPCC reporting guidelines⁶⁸. CH₄ and N₂O emissions can be reduced through marginal abatement cost curves⁶⁹. These estimates are derived based on available mitigation technologies in the short term, and extended for long-term mitigation scenarios to account for technological learning and removal of implementation barriers. For methane from enteric fermentation—the major CH₄ emission source—a maximum reduction potential of 29–50% depending on world region is estimated for 2050, rising to 35–60% in 2100. Maximum reductions of N₂O emissions from fertilization—the major agricultural N₂O source—range from 21% to 35% in 2050 and from 26% to 40% in 2100. The economic potential can also be smaller, depending on the level of the greenhouse gas price.

Scenario design. The business-as-usual scenario (BAU) follows the 'middle-of-the-road' storyline of the Shared Socioeconomic Pathways SSP2. The scenario set-up is well documented and results have been compared with other land system models⁷⁰. In this scenario, the total global income increases from US\$103 trillion to US\$231 trillion (05PPP) by 2050 (Supplementary Fig. 1), and average global per-capita income from US\$13,410 to US\$25,179 (05PPP) by 2050. Additionally, population increases from 8 billion people to 9.5 billion in 2070 before decreasing again to 9 billion by 2100.

The GDP-CAP scenario, which limits the high end of per-capita income and therefore the resulting food demand and dietary composition derived from past trends, reduces the per-capita income of all countries where per-capita income in 2020 is above US\$12,746 (05PPP) to this value by 2030. The income of US\$12,746 (05PPP) is the threshold between a middle- and high-income country as defined by the World Bank⁷¹. This value also corresponds roughly to the income level where mean life satisfaction begins to saturate^{72,73} and is furthermore very close to the current global average income of around US\$13,000 (05PPP) in 2020. Countries where per-capita income remains below this threshold continue to grow along the SSP2 baseline until they reach the threshold level. Total income decreases from US\$103 trillion to US\$77.7 trillion (05PPP) in 2030, although further growth in lower-income countries means that global income then peaks in 2080 at US\$119.3 trillion (05PPP). Global average per-capita income also decreases from US\$12,973 to US\$9,853 (05PPP) by 2035, and only increases to US\$12,696 (05PPP) by the end of the century as lower-income countries catch up to the sustainably scaled income (Supplementary Fig. 1).

The GDP-FAIR scenario implements the same reduction but concurrently implements a 'catch-up' of countries where per-capita income is below US\$12,746 (05PPP). These countries then see a linear increase of income until 2030 to US\$12,746 (05PPP) per capita. This amounts to a slight increase of total GDP from US\$103 trillion to US\$106 trillion (05PPP) by 2030 (as opposed to 144 US\$trillion (05PPP) in BAU), and due to population increase, incomes peak in 2070 at US\$120 trillion (05PPP) (Supplementary Fig. 1).

The DIET scenario implements a shift in consumer demand towards the planetary health diet as described in Willet et al.³⁶. The planetary health diet includes low amounts of animal-source foods, instead emphasizing fresh vegetables

and plant-based sources of proteins. For instance, calories from livestock products are limited to 200 kcal per person per day. The shift from current diets to the planetary health diet also takes place by 2030, for comparability with the income-based scenarios. Furthermore, we limit food waste to 20% of caloric food intake, which is approximately half of the current levels in high-income countries³⁶.

The EFF scenario follows the socioeconomic assumptions and dietary trends of the BAU scenario, but implements a GHG pricing on CO₂, N₂O and CH₄ emissions arising from LULUC and agricultural activities such as ruminant animal production. The price trajectory is calculated with REMIND-MAGPIE (ref. ⁷⁴), an integrated assessment modelling framework that arises from the coupling of the MAGPIE model to the REMIND energy-economy model⁷⁵. This enables the calculation of GHG prices to meet mitigation targets; here a price trajectory that limits warming to below 1.5°C in 2100 is used (globally uniform CO₂ price of US\$140 t⁻¹ in 2030 and US\$371 t⁻¹ in 2050; other GHGs are priced according to GWP100 CO₂ equivalent). By internalizing the external costs, the tax leads to a more efficient allocation of resources within the food system: area expansion into natural vegetation is substituted by intensification of existing areas, trading routes source commodities from less-polluting world regions and livestock production systems with lower emissions gain a competitive advantage. Moreover, a range of explicit mitigation technologies are implemented at additional costs to reduce emission factors⁶⁹, and forests are planted to sequester CO₂ (ref. ⁶³). Bioenergy demand was not changed in the mitigation scenario because the study is limited to the food system and cannot account for the emissions offset in the energy system by bioenergy.

The TRANS scenario is the combination of the scenarios GDP-FAIR, EFF and DIET. The dietary impacts of the GDP-FAIR scenario, however, overlap with the exogenous diet-shift assumptions of the DIET scenario; the inclusion of the GDP-FAIR assumptions thus solely ensures the consistency that people can also afford the sustainable diets.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

Generated data and replication scripts have been archived at: <https://doi.org/10.5281/zenodo.5543427>

Code availability

MAGPIE is an open-source model available at: <https://github.com/magpiemodel/magpie>. The model documentation for the exact version of MAGPIE used in this study (v.4.3.4) can be found at <https://rse.pik-potsdam.de/doc/magpie/4.3.4/>.

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Author contributions

B.L.B. and D.M.-C.C. designed the study and drafted the manuscript. B.L.B., D.M.-C.C., I.W., E.J.M.B. and H.L.-C. contributed to core model development. D.M.-C.C. undertook the modelling and analysis. All authors contributed to discussing the results and writing the paper.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to David Meng-Chuen Chen.

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Data collection The scenario data analysed in this paper are computed with the open-source framework MAgPIE (model version MAgPIE 4.3.4, doi.org/10.5281/zenodo.4730378), and can be replicated using the scripts archived at <https://doi.org/10.5281/zenodo.5888739>

Data analysis Data analysis was performed using R-libraries that are specifically built for processing MAgPIE inputs and outputs. These libraries are available open source at <https://github.com/pik-piam>, core libraries used are also archived: mrcommons (<https://doi.org/10.5281/zenodo.3822009>), mrland (<https://doi.org/10.5281/zenodo.3822083>), magpie4 (<https://doi.org/10.5281/zenodo.1158582>).

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